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Establishing the "Native Oxide Barrier Layer for Selective Electroplated" Metallization for Bifacial Silicon Heterojunction Solar Cells

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Abstract. The metallization of silicon heterojunction (SHJ) solar cells by selective Cu electroplating onto a structured PVD Cu-Al seed and mask layer stack is established by using the native oxide of the Al as insulating barrier. The NOBLE metallization (native oxide barrier layer for selective electroplating) allows reaching a first promising efficiency of 20.0% on full area SHJ solar cell with low contact resistivity on ITO. The approach features several advantages: low temperature processing, high metal conductivity of plated copper, no organic making and low material costs (almost Ag-free).

INTRODUCTION

The bifaciality of the crystalline silicon heterojunction (SHJ) architecture is quite attractive to increase the energy yield and lower the energy costs. However, the metallization costs for a standard screen-printed silver metal-grid on both sides of the SHJ solar cells become then more significant. Since the amorphous silicon (a-Si) or passivating layers are temperature sensitive, the standard silver metallization paste has to be adjusted for a low-temperature process [1,2]. These low-temperature (200°C) metal pastes are less conductive and a lot of efforts are actually dedicated to these metallization kind [3].

An attractive alternative metallization is the electroplating of highly conductive copper contacts already used in the printed electronics field [4]. Electroplating proved its potential with the outstanding efficiency of 25.1%, achieved on a large area bifacial SHJ solar cell [5]. The carrier selective junction (CSJ) solar cells are generally covered by transparent conductive oxide (TCO) as antireflective coating ARC and to promote the lateral conductivity on the passivating layers.

Plated metallization on TCO for solar cells faces several challenges [6–11]. While plated copper features low adhesion on TCOs, both-sides of the cell area are susceptible to plating. A metal seed layer deposited by physical vacuum deposition can promote the adhesion and serve as current distribution layer for plating. It can be deposited directly after vacuum deposition of the TCO on both sides of the solar cell, to allow bifacial plating. A standard metallization process – adapted from the microelectronics field, consists in plating the grid-positions after mask patterning of almost the full cell area with an organic resist. As illustrated in FIGURE 1 (a), the current distribution is not homogeneous without metal seed-layer leading to the half metallization of bifacial SHJ solar cells. On the contrary, the thin metal-seed layer deposited onto the TCO allows fast simultaneous bifacial electrodeposition of copper – FIGURE 1 (b). *CSEM* also demonstrated that high efficiencies above 24% are reached on bifacial full area SHJ solar cells with this metallization process [12].

However, the use of organic resist (consumables) covering almost 95% of the cell area – removed after plating, might be a hindrance for industrial implementation (resist cost, treatment of the water-wastes contaminated with organic products). Our established plated metallization concept aimed at reducing the process complexity and cost by avoiding the use of any organic masking to be even cheaper: low temperature, organic-free and almost Ag-free. A thin multifunctional metal stack with aluminum on top is deposited following the TCO deposition without

vacuum breaking. After structuration of the Al layer only in the grid-positions (≈ 5% of the cell area), selective Cu electroplating is performed on the thin metal-seed layer (below the Al) taking advantage of the self-passivation of the Al layer.

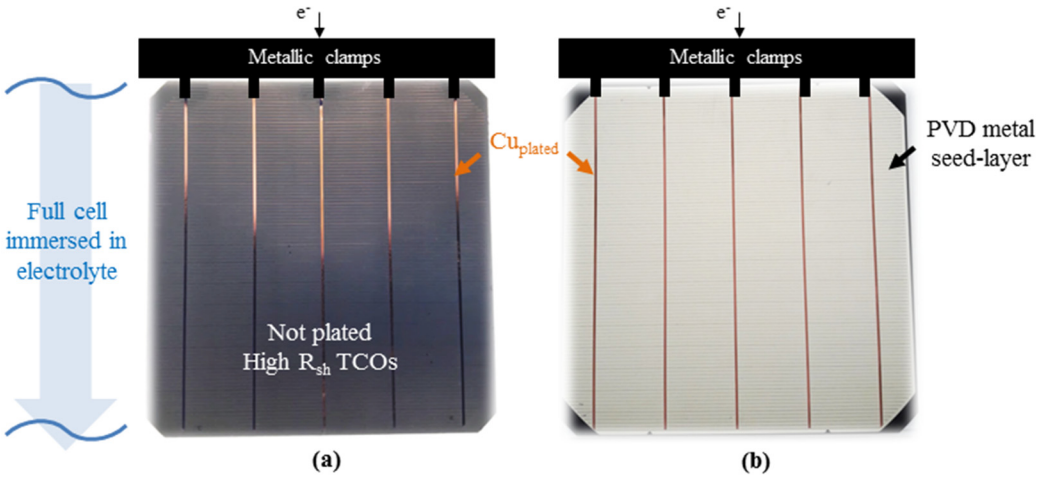

FIGURE 1. Pictures of the front side of two SHJ solar cells after full immersion in copper electrolyte for simultaneous bifacial plating of the contacts (fingers and busbars), (a) without metal seed-layer on the TCO and (b) with thin PVD metal seed-layer on the TCO.

APPROACH

Our novel *NOBLE* – "native oxide barrier layer for selective electroplated" metallization process is suitable for simultaneous bifacial plating onto SHJ or CSJ solar cells covered by TCOs [11]. The process development was carried on one batch of SHJ commercial precursors, covered on both-sides by indium-tin-oxide (ITO). This work focused on the optimization of several steps of the *NOBLE* metallization – depicted in FIGURE 2.

A thin PVD metal stack (Cu and Al on top < 200 nm thick) is sputtered following the TCO deposition – FIGURE 2 (1). The multifunctional physical vapor deposited (PVD) layers allow a homogeneous distribution of the current for simultaneous plating on both-sides of the solar cell as described above in FIGURE 1 (b). Furthermore, the self-passivation of the Al surface in ambient conditions is convenient for the selective plating of highly conductive Cu-contacts [9]. The grid-patterning (fingers and busbars) is then performed into the thin Al layer by chemical etching with inkjet-printing of an aqueous low concentrated caustic soda ($NaOH_{aq}$) ink in the desired grid positions – FIGURE 2 (2). The selective Cu electrodeposition is then achieved in the grid-positions onto the thin PVD Cu seed-layer – FIGURE 2 (3). Thin immersion silver capping (≈ 200 nm) is then deposited to protect the Cu-contacts from oxidation. The remaining PVD layers (Cu metal-seed and Al) in non-grid positions are then totally etched-back by chemical processing without damaging the underlying TCO – FIGURE 2 (4).

For early stage development only the front sides of the SHJ solar cells were metallized by the *NOBLE* process, first on small area after wafer dicing. However the PVD Cu-Al layers were deposited on both sides to allow I-V measurements. Besides the 1-sun current voltage (I-V) parameters of the solar cells measured under standard testing conditions (STC: AM1.5g, 100mW/cm²), contact angle measurements, optical and scanning electron microscopy (SEM) were also realized to characterize the process.

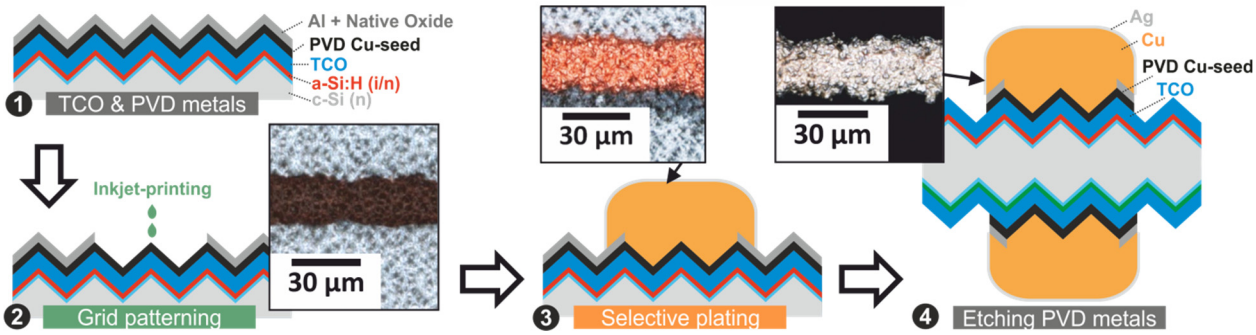

FIGURE 2. Sketch of the *NOBLE* metallization flow for bifacial SHJ solar cells and microscopic pictures of a finger (width < 30 µm) along the process. In step (1) the TCO and PVD metal-layers are deposited by PVD. (2) Inkjet-printing of a self-made low-concentrated $NaOH_{aq}$ ink follows to structure the Al layer in grid positions and (3) plate the Cu in the patterned areas simultaneously on both-sides of the solar cell. The PVD metal layers in non-grid positions are then chemically etched-back.

RESULTS AND DISCUSSIONS

The I-V parameters and the efficiencies of the lab-scale SHJ solar cells after *NOBLE* metallization are reported and plotted respectively in FIGURE 3. The I-V characteristics of the earliest mentioned solar cell metallized highlight several issues in the process flow at that time. The solar cell efficiency is limited by the short-circuit current J_{sc} (33.8 mA/cm²) among other characteristics. Indeed, the fingers metallized at that time were quite wide (> 100 µm). The other main limiting factor is the fill-factor FF of 70.7% lowered by finger interruptions and by an increase of the ITO sheet resistance R_{sh} after etchback of the PVD seed-layers (step (4) in FIGURE 2) due to chemical damaging. Finally, the open-circuit voltage V_{oc} is reduced by the small cell area [13] and by the damage of the passivation layers caused by a non-optimized sputtering.

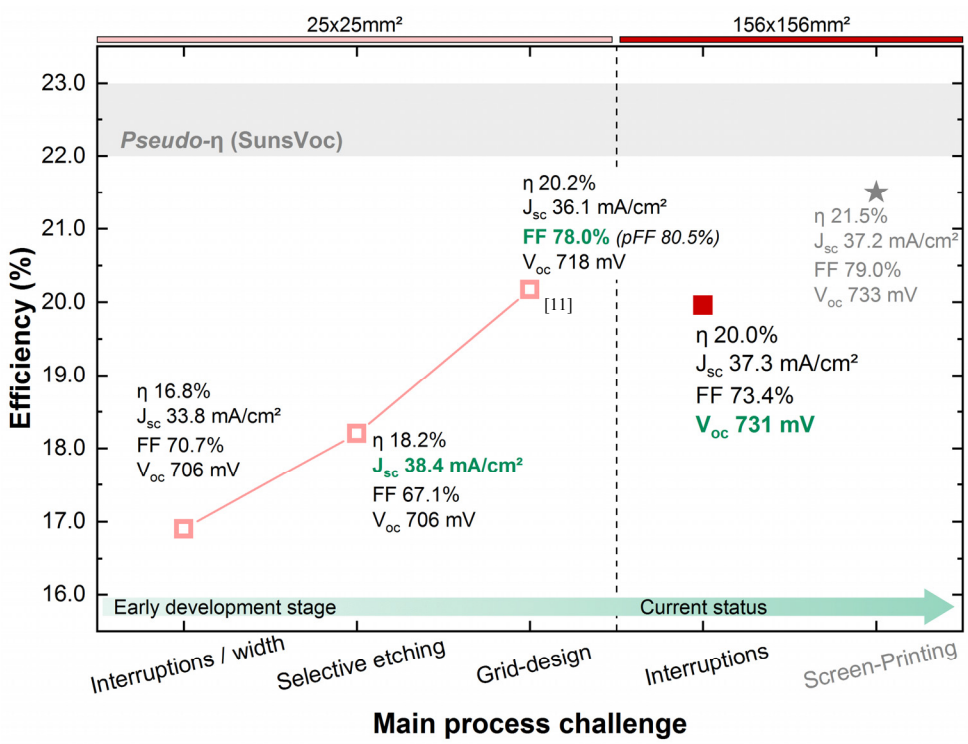

FIGURE 3. History of the *NOBLE* process development and I-V data on small area SHJ solar cells for the early stage development and on full cell area (156 x 156 mm²) for the current status.

Fine Line Structuring in the Al-Layer

Inkjet-printing of the low concentrated $NaOH_{aq}$ solution is impacted by the ink composition but also by the wetting properties between the water-based ink and the Al surface. To create fine lines in the Al-layer a hydrophobic wetting is suited, but this has to be weighed against the risk of discontinuities in the lines since the droplet volume is very low (2-5 pL). The PVD metal stack was deposited on the ITO covering the textured SHJ cells (random pyramids 4-7 µm) with sputtering or electron beam evaporation. During inkjet-printing of the $NaOH_{aq}$ solution, more or less spreading was observed, depending on the Al-layer deposition technique and history. The surface nanostructure of the samples was characterized by SEM before (FIGURE 4 (a)) and after (FIGURE 4 (b-c)) metal deposition. The thin ITO layer (≈ 75 nm) is polycrystalline with a random dendritic grain size around 20-30 nm. After metal stack deposition (Cu_{100nm} / Al on top), the nanostructure of the amorphous Al surface is flatter (FIGURE 4 (b), Al_{50nm}). The deposition of more Al – 100 nm, changes again the nanostructures to a porous rounder surface. The required quantity of $NaOH_{aq}$ ink to structure the Al-layer is higher for 100 nm but these different nanostructures did not show a significant impact on the spreading during inkjet-printing.

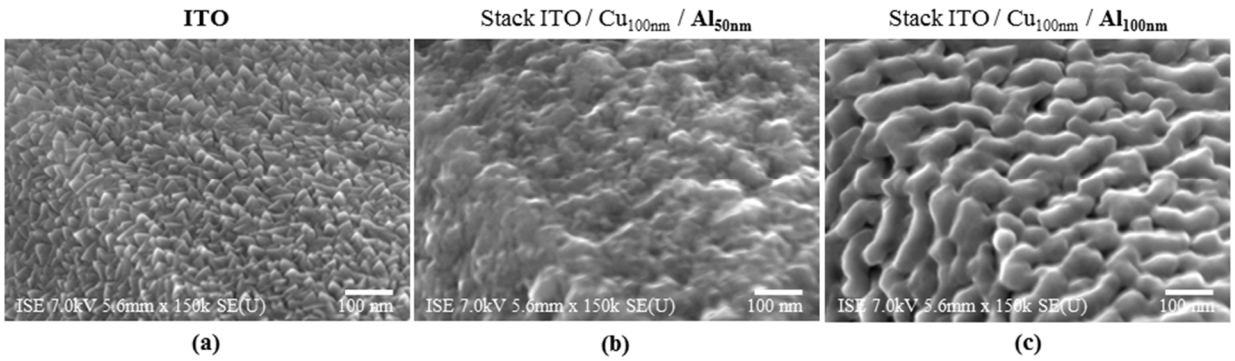

FIGURE 4. SEM pictures of the SHJ solar cell surface and nanostructures on a pyramid flank covered by (a) ITO and after metal sputtering on the ITO of (b) Cu_{100nm} / Al_{50nm} and (c) Cu_{100nm} / Al $_{100nm}$.

The wetting properties on the Al surface might also be impacted by its chemical composition. The wetting behaviour on the Al layer after different durations of air-exposure was studied by contact angle measurements (CA) with deionized water. The etching reaction of Al with low concentrated $NaOH_{aq}$ does not start instantaneously, so wetting properties should play a role in line width. After PVD deposition, the Al surface is well hydrophilic (contact angle – CA ≈ 30°). The samples were stored at ambient atmosphere (temperature of 23°C and humidity of 35%) to allow growth of the dense amorphous native oxide layer Al_2O_3 (self-passivation of Al). Twenty hours are sufficient for the surface to become less hydrophilic (CA ≈ 90°). Almost five days allow reaching a hydrophobic surface (CA ≈ 100-120°) which stabilizes after that time. The Al surface might be covered by a porous $Al(OH)_3$ which grows over the time depending of the storage temperature and humidity and decrease after the bonding capacity of the surface [14]. However, a drier storage atmosphere (50°C, 6% humidity) shows that the hydrophobic surface is reached already after one day (CA > 100°C). The native oxide and hydroxide growing at the Al-surface after PVD seem to be a key-factor to obtain a hydrophobic surface suitable for fine line etching in the Al layer with $NaOH_{aq}$ inkjet-printing. Pre-treatments of aluminum surfaces to instantaneously obtain and strengthen this behavior are currently under investigation.

Following this principle, a second SHJ solar cell was metallized by first patterning fine lines (20-30 µm wide) in the thin Al layer as observed in the microscopic picture in FIGURE 2 (2) where the underlying Cu-seed layer is visible. The following selective plating and etching steps were then performed. As reported and plotted in FIGURE 3, a better efficiency was reached due to a significant increase in J_{sc} to 38.4 mA/cm². Indeed, the shadowing at the SHJ solar cells was significantly reduced with the fingers below 30 µm wide – FIGURE 2 (3-4). However, finger interruptions occurred in the printing process. In addition, parasitic etching of the TCO upon etching of the PVD metal layer was also observed. Both effects still limit the FF of this cell.

Selective Etching of the PVD Metal Layers

Selective etching already allows the Al-patterning with $NaOH_{aq}$ without impacting significantly the underlying Cu-seed. This selectivity was also investigated to remove the thin PVD layers (Cu / Al) in the non-grid positions without affecting the plated contacts and the ITO. Prior to etching, a thin immersion silver layer (≈ 200 nm) is deposited on the Cu to protect the contact from oxidation. Our self-made $NaOH_{aq}$ ink can then also be used to etch the Al layer in non-grid positions as already described in another publication [11]. The Cu-seed can then be removed in an acidic solution of ammonium persulfate $(NH_4)_2S_2O_8$.

This further optimization of the *NOBLE* metallization enables to reach a first promising efficiency of 20.2% with low contact resistivity to ITO as observed in FIGURE 3. No finger interruptions were noticed explaining the good FF of 78.0% (pFF limited to 80.5% due to small area SHJ solar cell with large rim fraction [13]). However, the metal-grid design and the metal sputtering were not optimized explaining the remaining difference with the pseudo-efficiency simulated with SunsVoc measurements (lowered V_{oc} and pFF).

"NOBLE" Metallization on Full Area SHJ Solar Cells

Full area SHJ solar cells were metallized after this first process optimization of the *NOBLE* metallization on small area cells. As reference on 156 x156 mm² cells, SHJ precursors of the same batch were also metallized by the standard screen-printing technology of low-temperature Ag-paste. The highest efficiency reaches 21.5% with a moderate FF of 79.0%, still significantly below the pseudo-efficiency (FIGURE **3**). The *NOBLE* metallization of a first full area SHJ solar cell reaches an encouraging efficiency of 20.0% with a V_{oc} up to 731 mV meaning that the sputtering of metal did not impact the passivation quality and the rim effect could be overcome as expected. As mentioned earlier, the optimization of the inkjet-printing process should prevent finger interruptions and attain the full potential of these SHJ precursor solar cells around 22%.

CONLUSION AND OUTLOOK

Our metallization by the novel developed *NOBLE* approach – i.e. native oxide barrier layer for selective electroplating, creates advanced plated copper contacts for solar cells with TCO layers at greatly reduced process complexity. A first promising efficiency of 20.0% was established by this route on a full area SHJ solar cell. Optimizing Al-patterning by inkjet-printing of a self-made low concentrated $NaOH_{aq}$ ink to prevent finger interruptions is ongoing to reach the full potential of the SHJ solar cell technology.

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